

Supplementary Information

ReMol: A Chemical Reaction Knowledge-guided Self-supervised

Molecular Image Representation Learning Framework

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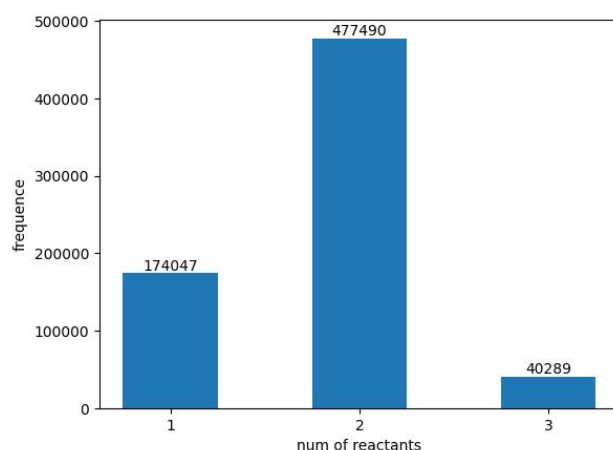
Supplemental Materials and Methods

A. Supplementary Experiment Setup

A.1 Dataset details

Pretraining Dataset

To pretraining ReMol, we first processed 1,013,118 reactions from USPTO-granted patents in the USPTO-full¹ dataset. We filtered the reactions that could not extract reaction templates and reactions that appeared in the downstream task test set to prevent data leakage. Finally, we obtained 691,826 reactions and divided them into a training set (80%) and a validation set (20%). As shown in the **Supplementary Figure 1**, we display the number of reactants in the pretraining dataset.



Supplementary Figure 1. Statistics of the number of reactants in the pretraining data.

Fine-tuning Dataset

As shown in the **Supplementary Table 1**, we display the details of the fine-tuning dataset.

Supplementary Table 1. Details of fine-tuning dataset.

Task	Dataset	Original download link	Description
Reaction Product Prediction	USPTO-479k	MolR	The dataset contains 478,612 chemical reactions, divided into training, validation, and test sets, containing 408,673, 29,973, and 39,966 reactions respectively.
Chemical Reaction Classification	Schneider	DRFP	The dataset contains 38,800 chemical reactions, divided into training, validation, and test sets, containing 31,002, 3,896, and 3,902 reactions respectively.
Molecular Property Prediction	MoleculeNet	KPGT	These datasets, eight are designed for classification tasks and the remaining three are designed for regression tasks.
Activity Cliff Estimation	MoleculeACE	MoleculeACE	Each cluster was then split into a training set (80%) and a testing set (20%) using stratified random sampling based on the activity cliff label.

A.2 Hyperparameters

Following to previous work^{2,3}, we use grid search to find the best combination of hyperparameters on downstream tasks. The specific search spaces are shown in **Supplementary Table 2**.

Supplementary Table 2. Hyperparameters for pretraining and fine-tuning of ReMol.

Hyperparameters	Pretraining	Fine-tuning
Max Epochs	50	[60, 100]
Batch Size	128	[8, 16, 32, 64]
Temperature	0.05	-
Learning Rate	5e-4	[5e-5, 5e-4, 3e-4, 5e-3]
Weight Decay	1e-5	[1e-4, 1e-5]
Momentum	0.9	0.9
Dropout Rate	0.1	[0.1, 0.5]
Learning Rate Decay	Cosine decay	Cosine decay
Image Size	224×224	224×224
Activator	-	[LeakyReLU, GELU]

B. Supplementary Methods

B.1 Time Complexity

The overall runtime of our method can be divided into two main components: data pre-processing and model computation. In summary, the time complexity analysis of the proposed method is shown in **Supplementary Table 3**.

Data Preprocessing: The primary computational cost of data preprocessing involves masking the reactant and product images and calculating the reaction fingerprint.

- ✓ **Masking:** The masking operation depends on the number of atoms and bonds in the reactants and products. Assuming the mask operation itself is $O(I)$, the time complexity of masking is $O(N \times (N_1 + N_2 + N_3 + N_4))$ where N_1 , N_2 , N_3 , and N_4 represent the number of atoms and bonds in the reactants and products, respectively, and N is the number of reactions. Assuming the number of atoms

and bonds in the reactants and products are approximately equal, this simplifies to $O(N \times M)$, where M represents the average number of atoms and bonds.

- ✓ **Reaction Fingerprint:** The time complexity of calculating the reaction fingerprint is related to the number of atoms, which is $O(N \times (N_1 + N_3))$, and similarly simplifies to $O(N \times M)$.

Model Calculation: The model computation complexity is divided into two parts: embedding calculation and pretraining task calculation.

- ✓ **Embedding Calculation:** The model consists of ResNet18 and Transformer components. The time complexity of ResNet18 depends on the input image size and is $O(H \times W \times N \times K^2 \times C^2)$, where H and W are the height and width of the input image, K is the kernel size, and C is the channel. Typically, K and C are constants, so the complexity simplifies to $O(H \times W \times N)$. The time complexity of the Transformer depends on the embedding dimension and is $O(N' \times N' \times D \times N)$, where N' is the sequence length (which can be ignored, as each reactant and product is treated as a token), and D is the embedding dimension. Thus, the time complexity simplifies to $O(N \times D)$.
- ✓ **Pretraining Task:** This paper proposes three pretraining tasks: MRCL, RPCL, and RRCL. The time complexity of MRCL is primarily within the preprocessing and embedding calculation steps, so its complexity is $O(1)$ in the pretraining task calculation. For RPCL, the primary computational cost is the calculation of similarity between all sample pairs in the batch during contrastive learning. The time complexity of this operation is $O(B^2 \times D)$ where B is the batch size. Similarly, RRCL also has a time complexity of $O(B^2 \times D)$.

Supplementary Table 3. The time complexity in the data preprocessing and model calculation, where N is the number of reactions, M represents the average number of atoms and bonds, H and W are the height and width of the input image, D is the embedding dimension and B is the batch size.

Operation	Time Complexity
Data Preprocessing	
Masking	$O(N \times M)$
Reaction Fingerprint	$O(N \times M)$
Model Calculation	
Embedding Calculation	$O(H \times W \times N) + O(N \times D)$
Pretraining Task	$O(B^2 \times D)$

B.2 Computational Requirements

The computational requirements of the proposed method are shown in

Supplementary Table 4.

Supplementary Table 4. The computational requirements in the pretraining and fine-tuning stages. #samples represent the total number of molecules.

(a) The computational requirements in the pretraining stage.				
#samples	#batch	GPU memory	Training time	Server
690K	128	~40G	~1 hours/epoch	CPU: 13th Gen Intel® Core™ i7-13700K GPU: 4090 Ti
(b) The computational requirements in the fine-tuning stage.				
#samples	#batch	GPU memory	Training time	Server
1000	16	2.7G	~0.5 min/epoch	CPU: 13th Gen Intel® Core™ i7-13700K GPU: 4090 Ti

C. Supplementary Results

C.1 Results on Pretraining

In the MRCI task, there are four parts of the loss, namely, mask reactant atom center identification L^R_{atom} , mask reactant bond center identification L^R_{bond} , mask product atom center identification L^P_{atom} , and mask product bond center identification L^P_{bond} . Finally, the loss L of MRCI is as follows: $L = L^R_{atom} + L^R_{bond} + L^P_{atom} + L^P_{bond}$

Finally, the loss of MRCI on the training set is 0.266, and the accuracy on the validation set is shown in **Supplementary Table 5**.

Supplementary Table 5. The accuracy of MRCI task. MRACI represents mask reactant atom center identification task.

Task	MRACI	MRBCI	MPACI	MPBCI
Acc	0.900	0.804	0.953	0.889

C.2 Statistical Significance Analysis

We have conducted additional statistical significance analyses to validate the robustness of our model’s improvements (See **Supplementary Table 6**). ReMol achieves statistically significant differences compared to baselines across all tasks ($p < 0.05$).

Supplementary Table 6. Statistical significance tests of ReMol. We performed significance tests on all four downstream tasks. For molecular property prediction and activity cliff estimation, we randomly selected one dataset.

Dataset	Baseline	t-test	U-test	Significant
USPTO-479k	MolR	$<< 0.05$	$<< 0.05$	yes
Schneider	MolR	0.001	0.001	yes
Estrogen	KPGT	0.046	0.031	yes
FreeSolv	KPGT	0.044	0.045	yes
D4R	ImageMol	0.025	0.030	yes

C.3 Dataset Bias Analysis

It is worth clarifying that USPTO-full primarily assesses model generalizability over known reaction types as a benchmark dataset originally designed for single-step retrosynthesis evaluation. Unlike conventional works that directly use it for retrosynthesis prediction, our study innovatively employs it as a pretraining corpus to extract general chemical knowledge (e.g., reaction center) through self-supervised learning. As correctly noted, the absence of explicit reaction type labels (e.g., SN2, Suzuki coupling) in USPTO-full does limit direct quantitative analysis of reaction type distribution.

To rigorously investigate potential bias impacts, we conducted systematic experiments on the Schneider dataset (chemical reaction classification task) by constructing controlled imbalance scenarios. Given its balanced original reaction type distribution, we simulated artificial imbalance by retaining only 10% and 5% of samples from the first five reaction categories while keeping all 41 remaining categories intact. This setup tests whether models disproportionately favor majority reaction types. As shown in **Supplementary Table 7**, both 10% and 5% imbalanced classes caused notable performance degradation (1.3% and 3.2% overall accuracy drops respectively), particularly in imbalanced categories (9.7% and 25% accuracy decreases). These results confirm that insufficient representation of specific reaction types during training indeed compromises model performance, highlighting the necessity for comprehensive pretraining datasets encompassing diverse reaction types.

Supplementary Table 7. Class U_1 represents the imbalanced class (Sampled Class: 0,1,2,3,4) with each class having only 10% of the full class. Class U_2 represents extremely imbalanced class (Sampled Class: 0,1,2,3,4), with each class having only 5% of the full class. Acc_imbalanced indicates the accuracy of the imbalanced class (Sampled Class: 0,1,2,3,4). The accuracy is computed multiple times on different random samples, and we report the mean and standard deviation results on the test set.

Reaction class in training set	Acc	Acc_imbalanced
Full	0.945(0.001)	0.952(0.001)
5 U_1 classes + 41 Full classes	0.933(0.002)	0.860(0.001)
5 U_2 classes + 41 Full classes	0.915(0.004)	0.714(0.008)

C.4 Few-shot Learning Analysis

We show the performance of each pretraining task under few-shot conditions in the Extended Table 8. We attribute this strength of ReMol's small reaction classification dataset capability to two synergistic factors: large-scale pretraining on nearly one million reactions transfers fundamental chemical reaction pattern recognition, while our reaction-aware pretraining tasks, especially Reaction Representation Contrastive Learning (RRCL), distill discriminative reaction-level semantics. RRCL constructs comprehensive reaction fingerprints that integrate both reactant features and product transformations, capturing critical reaction-level features. Crucially, through contrastive learning, RRCL forces the model to cluster chemically similar reactions in the embedding space. This capability enables robust classification even in data-scarce scenarios, as evidenced by RRCL alone boosting small reaction classification dataset accuracy.

Supplementary Table 8. The analysis of ReMol for few-shot reaction classification.

Model	Non-Pretrain	+MRCI	+RPCL	+RRCL	ReMol
32	0.504(0.007)	0.697(0.003)	0.752(0.006)	0.788(0.009)	0.816(0.010)

C.5 Ablation Study

To further validate the effectiveness of our architecture, we conducted additional ablation studies by modifying the backbone network and adjusting the Transformer structure, as shown in **Supplementary Table 9**. Below are the key findings:

- ✓ **Replacing ResNet18 with ResNet50:** We replaced ResNet18 with ResNet50 while keeping the Transformer configuration unchanged (4 layers, 4 heads). Results showed a slight performance improvement, but the training cost increased significantly. Given the marginal gain and high computational overhead, we retained ResNet18 for efficiency and scalability.
- ✓ **Deepening the Transformer (4 → 8 layers/heads):** We increased the Transformer depth and heads from 4 to 8 while retaining ResNet18. Surprisingly, this led to performance degradation, likely because the Transformer is primarily employed to address the issue of inconsistent numbers of reactants and products in our task. Unlike natural language processing (NLP), where the number of tokens is much smaller than the sequence length, a deeper Transformer does not necessarily extract the reaction information in our case. This suggests that a shallower Transformer (4 layers/heads) better balances model capacity and generalization in our task.

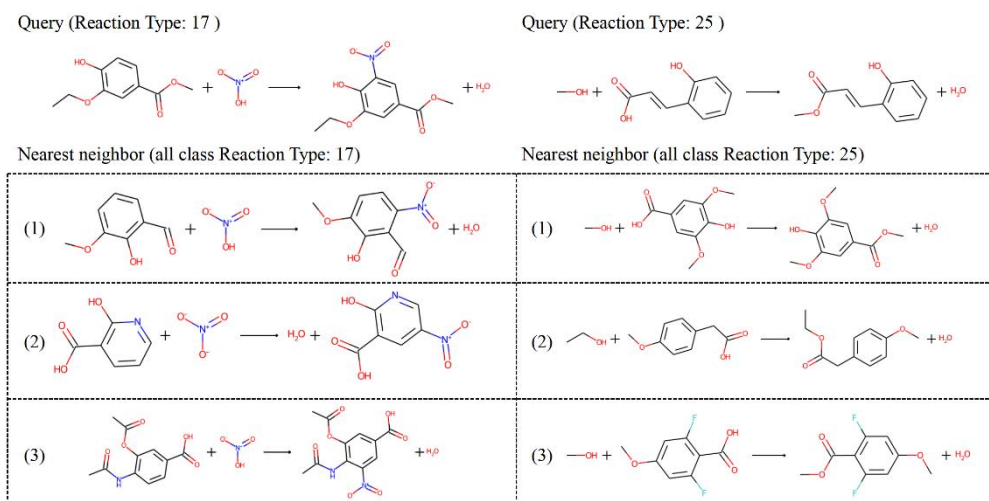
Supplementary Table 9. Ablation results of ReMol with different network components on molecular property classification and regression datasets. The bold indicate the results in our original paper.

Model	Classification dataset	Regression dataset
	AVG	AVG
ResNet18+Transformer (with 4 layers and 4 heads)	0.847	1.137
ResNet50+Transformer (with 4 layers and 4 heads)	0.850	1.139
ResNet18+Transformer (with 8 layers and 8 heads)	0.821	1.245

C.6 Case Study

Similar Reaction Search

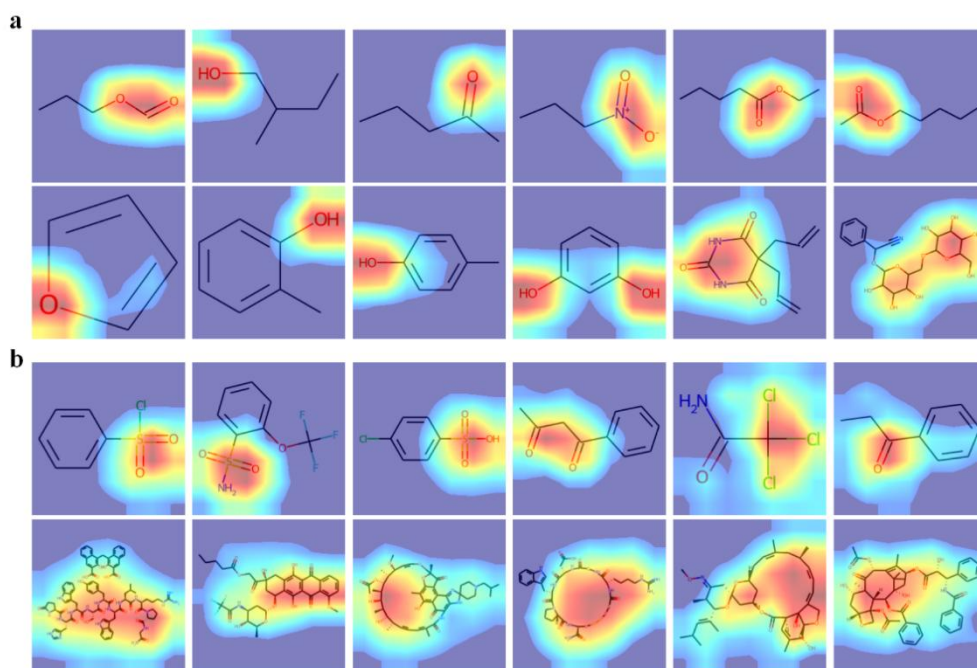
Supplementary Figure 2 shows the top3 nearest neighbors found in the test set. All of the retrieved neighbors are of the same class as the query reaction and share the same reaction template. This means that ReMol can utilize template information for reaction classification.



Supplementary Figure 2. A similar reaction search using ReMol features.

Feature Contribution

We conducted additional experiments using Grad-CAM to visualize the attention of the ReMol model (see **Supplementary Figure 3**). Our analysis revealed critical insights into the model's decision-making process, aligning with domain-specific chemical principles: (1) The model pays significant attention to polar groups such as hydroxyl (-OH) and carboxyl (-COOH) in the ESOL dataset (**Supplementary Figure 3 a**). This is highly consistent with chemical intuition⁴, because such groups significantly affect the water solubility of molecules through hydrogen bonding. (2) The attention mechanism successfully captures known toxicity-related structures⁵ (e.g., benzene rings, macrocyclic compounds) as well as detoxifying groups (e.g., halogen atoms, carboxyl groups) (**Supplementary Figure 3 b**).



Supplementary Figure 3. Attention interpretation of ReMol. (a) Examples of ReMol's heatmaps of ESOL dataset by Grad-CAM. (b) Examples of ReMol's heatmaps of Tox21 dataset.

Supplementary References

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